

IT Infrastructure Database: ERD Documentation

1. Overview

This database is designed to manage IT infrastructure, tracking physical hardware (Hosts), Virtual Machines (VMs), IP addressing, user ownership, and security vulnerabilities. This aligns with the core requirements of the semester project to build an integrated infrastructure management system.

2. Design Decisions & Assumptions

- **Host vs. VM Separation:** Physical hosts (HOSTS) and virtual machines (VIRTUAL_MACHINES) are modeled as separate entities. This accurately reflects infrastructure topologies where multiple VMs reside on a single physical host (`host_id` as a Foreign Key in the VMs table).
- **IP Allocation:** A central IP_ALLOCATIONS table manages individual IP addresses, linked to a broader NETWORK_SEGMENTS table. A Host is assigned exactly one primary IP (1:1 relationship via `ip_id`), assuming hosts only need tracking for their primary management interface for this MVP phase.
- **Ownership:** VMs are tracked by user ownership (`owner_user_id`), assuming physical hosts are managed globally by the IT team, while individual VMs are provisioned to specific users.
- **Vulnerability Tracking:** Vulnerabilities (VULNERABILITIES) are mapped through an associative/junction table (SCAN_RESULTS). This resolves the many-to-many relationship, recording specific instances of a vulnerability found on a specific Host or VM on a given date.

3. Normalization

The schema has been designed to meet Third Normal Form (3NF) requirements:

- **1NF:** All tables have a designated primary key. All attributes contain atomic values.
- **2NF:** All non-key attributes are fully dependent on the primary key. For example, `username` depends entirely on `user_id`.
- **3NF:** There are no transitive dependencies. For instance, instead of storing the `department_name` in the USERS table, we store a `department_id` which references

the DEPARTMENTS table.

4. Cardinality & Relationships

- **DEPARTMENTS to USERS (1:M):** One department has many users.
- **USERS to VIRTUAL_MACHINES (1:M):** One user can own many VMs.
- **NETWORK_SEGMENTS to IP_ALLOCATIONS (1:M):** A segment contains many IP addresses.
- **IP_ALLOCATIONS to HOSTS (1:1):** An IP address is assigned to one host.
- **OPERATING_SYSTEMS to HOSTS/VMs (1:M):** One OS type can be installed on many systems.
- **HOSTS to VIRTUAL_MACHINES (1:M):** A single host runs multiple VMs.
- **HOSTS/VIRTUAL_MACHINES to SCAN_RESULTS (1:M):** A system can have multiple scan results.
- **VULNERABILITIES to SCAN_RESULTS (1:M):** A single CVE can appear in many scans.